

Dynamics:- Dynamics is the branch of mechanics which deals with the study of motion of a particle. Dynamics is divided into two parts.

- i) Kinematics and
- ii) Kinetics.

i) Kinematics:- It is the branch of dynamics which deals with the study of motion of a body or particle without the effect of forces causing motion. It deals with the study of displacement, velocity and acceleration.

ii) Kinetics:- It is the study of motion of bodies and its relationship with time considering the forces causing the motion. It deals with the means of determining the forces required to produce such motion.

Types of kinematics:- Depending upon type of motion kinematics is subdivided into rectilinear motion and curvilinear motion.

a) Rectilinear motion:- The motion of a particle on the body along the

straight path is called rectilinear motion.

Eg: i) Path traced by a stone when dropped from certain height.

ii) Path traced by a moving car along the straight line.

b) Curvilinear motion:- The motion of a particle or the body along the curved path is called as curvilinear motion.

Eg: i) Path traced by a bomb dropped from an aeroplane.

ii) Path traced by a stone thrown in 15 space.

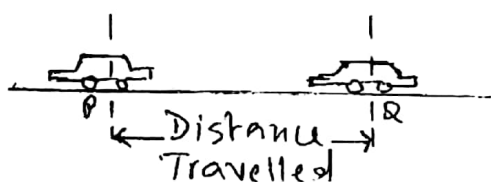


Fig: Rectilinear motion.

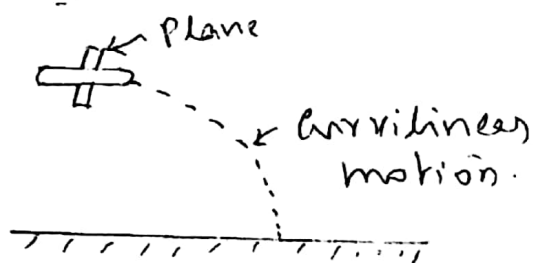


Fig: Curvilinear motion.

Definition:-

Displacement:- The position occupied by a particle moving along a straight line at any particular instant of time 't' from its initial position is called displacement.

Distance travelled (s):- It is the total distance moved by a body. If a body is moving with an uniform velocity 'v', then in 't' secs, the distance travelled mathematically is, $s = vt$.

Speed:- The speed of a body is the rate of change of its position with time irrespective of its direction and is thus a scalar quantity.

$$\text{Speed} = \frac{\text{Distance moved}}{\text{Time interval}}$$

Velocity:- It is the rate of change of displacement in a specific direction. It is a vector quantity.

Average velocity:- The initial velocity of a particle may be 'v₁' at time 't₁' and final velocity is 'v₂' at time 't₂'. The average velocity $v_{\text{avg}} = \frac{v_1 + v_2}{2}$.

$$\text{or } v_{\text{avg}} = \frac{\Delta s}{\Delta t}$$

Acceleration:- It is the rate of change of velocity with respect to time.

Acceleration = $\frac{\text{Change of velocity}}{\text{Time interval}}$

$$a = \frac{v_2 - v_1}{t} \text{ m/sec}^2.$$

Uniform acceleration:- The body moving in such a way that its velocity changes equal in magnitude at equal intervals of time is said to be in uniform acceleration.

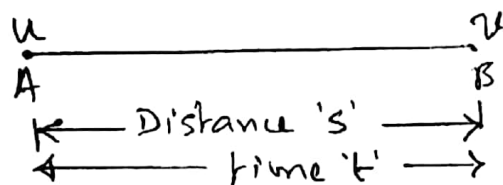
Variable acceleration:- If a body moves in such a way that, its velocity changes unequal in magnitude in equal interval of time, it is said to be moving with a variable acceleration.

Instantaneous velocity:- Instantaneous velocity of a particle at any instant 't' is obtained as,

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}.$$

Newton's Laws of motion:- Already discussed in the previous chapter.

Equations of motion with uniform Acceleration:-



Consider a body moving in a straight line, A to B with uniform acceleration.

Let u = Initial velocity of body at 'A'.

v = Final velocity of body at 'B'.

a = Uniform acceleration.

t = Time taken to move from A to B.

s = Distance travelled from A to B.

According to definition

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{time.}}$$

$$a = \frac{v - u}{t}$$

$$\text{or } \boxed{v = u + at} \quad \text{--- (1)}$$

$$\begin{aligned} \text{Also distance} &= \text{average velocity} \times \text{time} \\ &= \left(\frac{u + v}{2} \right) \times t \end{aligned}$$

Substituting value of 'v' from (1)

$$s = \left(u + \frac{u+at}{2} \right) t$$

$$2s = \cancel{2ut} + \cancel{at^2}$$

$$\boxed{s = ut + \frac{1}{2} at^2} \dots (2)$$

Also from equation (1)

$$t = \left(\frac{v-u}{a} \right)$$

Substituting the value of 't' in (2)

$$s = u \left(\frac{v-u}{a} \right) + \frac{1}{2} a \left(\frac{v-u}{a} \right)^2$$

$$= \frac{uv - u^2}{a} + \frac{1}{2} a \left[\frac{v^2 + u^2 - 2uv}{a^2} \right]$$

$$= \frac{uv - u^2}{a} + \frac{v^2 + u^2 - 2uv}{2a}$$

$$= \frac{2uv - 2u^2 + v^2 + u^2 - 2uv}{2a}$$

$$= \frac{v^2 - u^2}{2a}$$

$$2as = v^2 - u^2$$

$$\boxed{v^2 - u^2 = 2as} \dots (3)$$

The equations ① ② and ③ are called equations of rectilinear motion.

Acceleration due to Gravity:-

A body which is free to move entirely under the influence of the earth's gravitational attraction will be subjected to an acceleration directed towards the centre of the earth. This acceleration is called the acceleration due to gravity. For uniform acceleration its value is generally taken as $g = 9.8 \text{ m/sec}^2$.

Solved examples:- (Rectilinear motion)

→ A car starts with a velocity of 4 m/sec and moves in a straight line with constant acceleration. If its velocity at the end of 5 seconds is 6.5 m/sec , determine,

- i) The uniform acceleration
- ii) Distance travelled in 5 seconds.

Solution: -

$$u = 4 \text{ m/s}, \quad v = 6.5 \text{ m/s}, \quad t = 5 \text{ sec}$$

We have

$$v = u + at$$

$$6.5 = 4 + a \times 5$$

$$\boxed{a = 0.5 \text{ m/s}^2}$$

Distance travelled in 15 sec, $t_2 = 15 \text{ sec}$

$$\begin{aligned} s &= ut + \frac{1}{2} at^2 \\ &= 4 \times 15 + \frac{1}{2} \times 0.5 \times 15^2 \\ &= \underline{\underline{116.25 \text{ m}}} \end{aligned}$$

2) A car starts from rest and accelerated uniformly. It a speed of 75 kmph. over a distance of 1000m. Find the acceleration of the car and time taken to attain this speed.

If a further acceleration rises to the speed to 100 kmph. in 10 sec; find new acceleration and further distance moved.

Solution:-

Case (i)

$$u = 0$$

$$v = \frac{75 \times 1000}{60 \times 60} = 20.833 \text{ m/sec}$$

$$s = 1000 \text{ m}$$

We have.

$$v^2 - u^2 = 2as$$

$$20.833^2 - 0^2 = 2 \times a \times 1000$$

$$a = \underline{0.217 \text{ m/sec}^2}$$

Also, $v = u + at$

$$20.833 = 0 + 0.217 \times t$$

$$t = \underline{96 \text{ sec}}$$

Case ii New acceleration and further distance moved in 10 sec to attain 100 kmph.

$$u = 20.833 \text{ m/sec} \quad v = \frac{100 \times 1000}{60 \times 60} = 27.777 \text{ m/sec}$$

$$v = u + at$$

$$27.777 = 20.833 + a \times 10$$

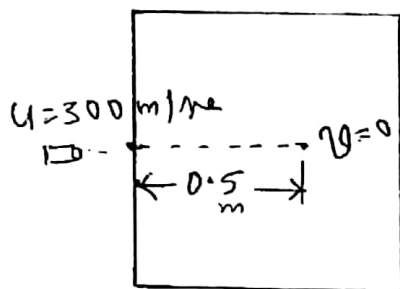
$$a = 0.6944 \text{ m/sec}^2$$

$$s = ut + \frac{1}{2}at^2 = 20.833 \times 10 + \frac{1}{2} \times 0.6944 (10)^2 = \underline{243.05 \text{ m}}$$

3) A bullet moving at the rate of 300 m/s is fixed in to a thick wood penetrates up to 500 mm . If it is fixed in to a 250 mm thick wood find the velocity of emergence. Take the resistance to be uniform in both the cases.

Solution

Case i) When the bullet is fixed in to 500 mm thick wood the velocity goes on decreasing to a value zero at 500 mm penetration.



Let a = Uniform retardation.

Initial velocity $u = 300 \text{ m/s}$

Final velocity $v = 0$.

Distance travelled $= s = 500 \text{ mm}$
 $= 0.5 \text{ m}$

We have $\boxed{v^2 - u^2 = 2as}$

$$0^2 - 300^2 = 2 \times a \times 0.5$$

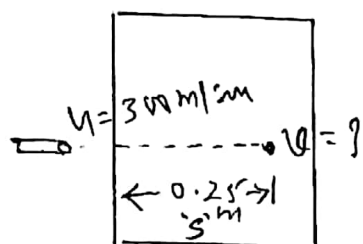
$$a = -90000 \text{ m/s}^2$$

Case ii) When the bullet is fixed in to 250 mm thick wood retardation remains same. $a = -90000 \text{ m/s}^2$

Initial velocity $= 300 \text{ m/s}$

Final velocity $v = ?$

Distance travelled $s = 250 \text{ mm} = 0.25 \text{ m}$.



We have, $v^2 - u^2 = 2as$

$$v = 2as + u^2$$

$$= 2 \times (-90000) \times 0.25 + 30^2$$

$$= \underline{\underline{212.132 \text{ m/s}}}$$

4) A car starts from rest with an acceleration of 5 m/s^2 . What is the distance travelled in 10^{15} seconds.

Solution:- Distance travelled in 10^{15} second.

$u = 0$ 9¹⁵ second 10¹⁵ second.

Distance travelled in 10^{15} second =

Distance travelled in 10^{15} second - Distance travelled in 9^{15} second.

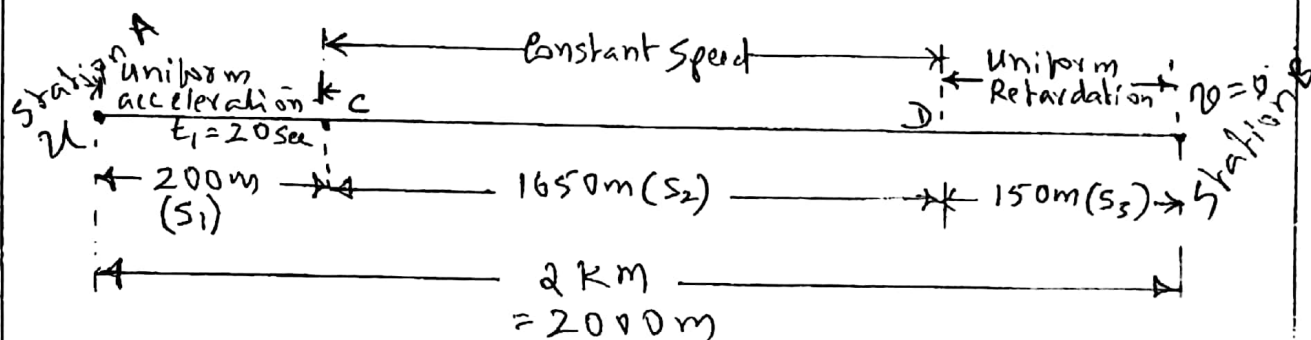
We have $s = ut + \frac{1}{2}at^2$

Distance travelled in 10th second

$$\begin{aligned}
 S &= \left[ut + \frac{1}{2}at^2 - ut + \frac{1}{2}at^2 \right] \\
 &= \left[0 \times 10 + \frac{1}{2} \times 5 \times 10^2 - 0 \times 10 + \frac{1}{2} \times 5 \times 10^2 \right] \\
 &= \underline{\underline{47.5 \text{ m}}}
 \end{aligned}$$

5) An ~~electro~~ electric train travelling between two stations 2.0 km apart, is uniformly accelerated for the first 20 seconds, during which it covers 200m. It then runs with a constant speed until it is finally retarded uniformly in the last 150m. Find the maximum speed of the train and the time taken to complete the journey between two stations.

Solution:



Consider motion of train from A to C

$u = 0$, $v_1 = \text{velocity at C}$ $t_1 = 20 \text{ sec}$, $s_1 = 200 \text{ m}$

We have, $s = ut + \frac{1}{2}at^2$

$$100 = 0 \times 20 + \frac{1}{2}a_1(20)^2$$

$$a_1 = 1 \text{ m/sec}^2$$

Using relation $v_1 = u + at$

$$= 0 + 1 \times 20$$

$$\boxed{v_1 = 20 \text{ m/sec}}$$

Consider the motion of train from C to D. (Uniform velocity)

$$v_1 = 20 \text{ m/sec} \quad s_2 = 1650 \text{ m}$$

$t_2 = \text{time taken from C to D}$

$$\boxed{s = v \times t}$$

$$1650 = 20 \times t_2$$

$$t_2 = 82.5 \text{ seconds}$$

Consider motion of train D to E (Uniform retardation)

Retardation:

$$u = v_1 = 20 \text{ m/sec} \quad v = v_3 = 0 \quad s = s_3 = 150 \text{ m}$$

$$\text{We have } v^2 - u^2 = 2as$$

$$0 - 20^2 = 2 \times a \times 150$$

$$a = -\underline{\underline{1.33 \text{ m/sec}^2}} \text{ (Retardation)}$$

Also using

$$v = u + at$$

$$0 = 20 + (-1.33) t_3$$

$$t_3 = \underline{\underline{15 \text{ sec}}}$$

$\therefore$ Maximum speed = 20 m/seconds.

$$\text{Total time of travel} = t_1 + t_2 + t_3$$

$$= 20 + 82.5 + 15$$

$$= \underline{\underline{117.5 \text{ sec}}}$$

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Using relation

$$v = u + at$$

$$0 = 20 + (-1.333) t_3$$

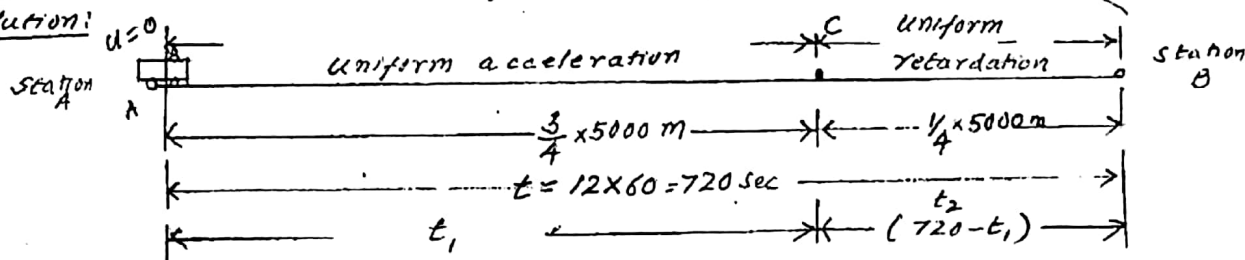
$$t_3 = 15 \text{ Seconds.}$$

Ans: Maximum speed of train = $20 \text{ m/sec} = 72 \text{ Kms/hr}$

Total time of travel = $t_1 + t_2 + t_3 = 20 + 82.5 + 15 = 117.5 \text{ Sec}$

Q.6 An electric train starts from a station and comes to another station which is 5 kilometers from the first on a straight track. For three quarters of the distance, the train is uniformly accelerated and for the remainder uniformly retarded. If the train takes 12 minutes to cover the whole journey, find (i) maximum speed the train attains; (ii) acceleration of the train; and (iii) retardation of the train.

Solution:



Consider motion of train from A to C. (Uniform acceleration)

initial velocity $u_a = 0$, final velocity $v_c = \text{maximum velocity}$

distance $AC = S_1 = 3750 \text{ m}$

time $= t_1$. Let a_1 : acceleration

using relation $v^2 - u^2 = 2as$

$$v_c^2 - 0 = 2a_1 \times 3750 \quad \therefore v_c^2 = 7500 a_1 \quad \text{--- (1)}$$

using relation $v = u + at$

$$v_c = 0 + a_1 t_1 \Rightarrow v_c = a_1 t_1 \quad \text{--- (2)}$$

Consider motion of train from C to B (Uniform retardation)

initial velocity $u = v_c$, final velocity $v = v_b = 0$

distance = 1250 m

time $t = (720 - t_1)$

retardation = a_2

using relation $v^2 - u^2 = 2as$

$$0 - v_c^2 = 2a_2 \times 1250 \Rightarrow v_c^2 = -2500 a_2 \quad \text{--- (3)}$$

using relation $v = u + at$

$$0 = v_c + a_2 (720 - t_1) \therefore v_c = -a_2 (720 - t_1) \quad \text{--- (4)}$$

Dividing eqⁿ (1) by (2)

$$\frac{v_c^2}{v_c^2} = \frac{7500 a_1}{-2500 a_2} \therefore \boxed{-a_2 = 3a_1} \quad \text{--- (5)}$$

Dividing (2) by (4) $\frac{v_c}{v_c} = \frac{a_1 t_1}{-a_2 (720 - t_1)} \Rightarrow \boxed{-a_2 (720 - t_1) = a_1 t_1} \quad \text{--- (6)}$

Substituting (5) in to eqⁿ (6) we get

$$3a_1(720 - t_1) = a_1 t_1$$

$$2160 - 3t_1 = t_1 \quad \therefore t_1 = 540 \text{ sec.} \quad t_2 = 720 - 540$$

$$\text{using relation } s_1 = ut_1 + \frac{1}{2} a_1 t_1^2 \quad t_2 = 180 \text{ sec.}$$

$$3750 = 0 + \frac{1}{2} a_1 (540)^2$$

$$a_1 = 0.02572 \text{ m/sec}^2 = \frac{0.02572 \times (3600)^2}{1000} = 333.33 \text{ km/hr}^2$$

$$a_2 = -3 \times a_1 = -3 \times 333.33 = 1000 \text{ km/hr}^2$$

$$\text{Maximum speed at C} \quad v_c = a_1 t_1 = 0.02572 \times 540$$

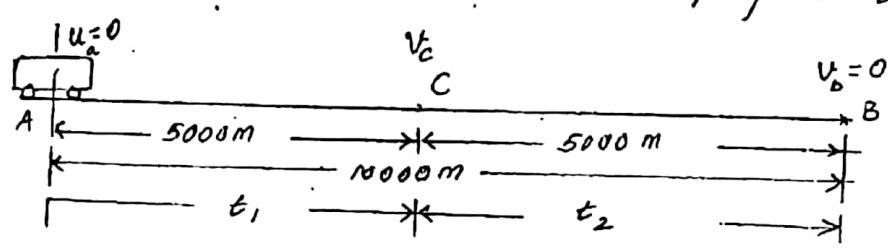
$$= 13.89 \text{ m/sec}$$

$$= \frac{13.89 \times 3600}{1000} = 50 \text{ kmph.}$$

Ex: 7

A bus travels between two stops 1000m. apart in 900 seconds. Assuming that its motion is with uniform acceleration for the first part of the journey and with uniform retardation for the next part, prove that the maximum velocity of the train is 80kmph.

Solution:



The vehicle moves from A to C with uniform acceleration over a distance of 500m and with initial velocity zero at A reaching final velocity v_c at C.

Similarly the vehicle moves from C to B with uniform retardation over a distance of 500m with initial velocity v_c and final velocity zero at B.

$\therefore$ acceleration from A to C = deceleration from C to B

$$\therefore t_1 = t_2 = \frac{900}{2} = 450 \text{ sec.}$$

$$s_1 = s_2 = 500 \text{ m}$$

Consider motion of vehicle from A to C. Using $s = ut + \frac{1}{2} at^2$

$$5000 = 0 \times 450 + \frac{1}{2} \times a (450)^2$$

$$\therefore a = 0.04938 \text{ m/sec}^2$$

$$\text{using relation } v_c^2 - u^2 = 2as$$

$$v_c^2 - 0 = 2 \times 0.04938 \times 5000$$

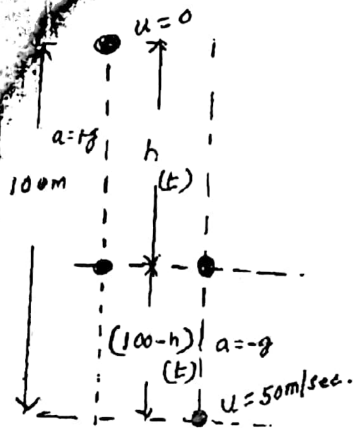
$$v_c^2 = 493.827 \quad \therefore v_c = 22.222 \text{ m/sec}$$

$$\text{or } v_c = \frac{22.222 \times 3600}{1000} = 79.999 \text{ m} \approx 80 \text{ m}$$

Hence proved.

(9)

stone is dropped from the top of a building 100 m high. At same time another stone is thrown upwards from the foot of building with a velocity of 50 m/sec. When and where the two stones cross each other.



Solution: Let h : height from top when two stones cross each other.

$(100-h)$: height from foot of the building when two stones cross each other.

Since both stones start moving at the same instant of time, when they cross each other let time taken be t .

acceleration for dropped stone $= a = +g$

acceleration for the upward moving stone $= a = -g$

Consider downward moving stone.

$$\text{Using } s = ut + \frac{1}{2} at^2$$

$$h = 0 \times t + \frac{1}{2} \times 9.81 t^2 = 4.905 t^2 \quad \dots \dots \dots (1)$$

Consider motion of upward moving stone: Using $s = ut + \frac{1}{2} at^2$

$$(100-h) = 50t + \frac{1}{2}(-9.81)t^2 \quad \dots \dots \dots (2)$$

Substituting 1 from eqⁿ (1) in to eqⁿ (2) we get

$$100 - 4.905 t^2 = 50t - 4.905 t^2$$

$$\therefore 50t = 100 \quad \therefore t = 2 \text{ seconds}$$

The two stones cross each other after 2 seconds.

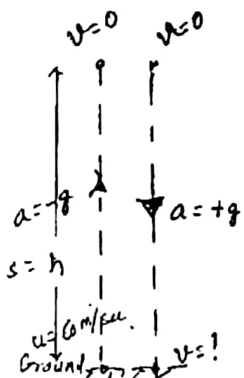
$$h = 4.905 \times 2^2 = 19.62 \text{ m}$$

The two stones cross each other at a height of 19.62 m from top or 80.38 m from bottom.

Ex: 4

A ball was thrown vertically upwards from the ground with the velocity of 60 m/sec. After 3 seconds another ball was thrown vertically upwards from the ground. If both the balls strike the ground at the same time, determine the velocity with which the second ball was thrown.

Consider motion of the first ball



Let t : time of travel where velocity reaches zero.

h : distance travelled in time t $a = -g$ (upward motion)

Using relation $v^2 - u^2 = 2as$

$$0 - 60^2 = 2 \times (-9.81)h \quad \therefore h = 183.486$$

$$\text{Using } v = u + at$$

$$0 = 60 - 9.81t \quad t = 6.116 \text{ Sec.}$$

Total time to reach the ground $= 2t = 12.232 \text{ Sec.}$

Consider motion of 2nd ball: The ball is thrown in to

space after 3 seconds the first ball is thrown, but they reach the ground at the same time.

Motion of 2 ball.

Motion of a particle under Gravity: When a particle is dropped in space it will be attracted by the gravitational force of earth and moves with uniform acceleration $g = 9.81 \text{ m/sec}^2$ in a vertical straight line. All equations derived previously holds good with replacement of a by g .

Case-1: when the particle is moving downwards $a = +g$

$$v = u + gt$$

$$v^2 - u^2 = 2gs$$

$$s = ut + \frac{1}{2}gt^2$$

Case-2: when the particle is moving upward against gravitational force

$$v = u - gt$$

$$v^2 - u^2 = -2gs$$

$$s = ut - \frac{1}{2}gt^2$$

with usual notations.

Ex:1 A stone is dropped from the top of a tower 50m above the ground. Find the velocity with which it strikes the ground and also time taken by the stone to reach the ground.

	$u = 0$ $a = g = 9.81 \text{ m/sec}^2$ $s = 50 \text{ m}$ $t = ?$ $v = ?$	Data: Initial velocity $u = 0$, final velocity $v = ?$ distance travelled $s = 50 \text{ m}$ $a = g = 9.81 \text{ m/sec}^2$ time $= t = ?$ Using equation $v^2 - u^2 = 2gs \Rightarrow v^2 = 2 \times 9.81 \times 50$ $\therefore v = 31.321 \text{ m/sec.}$
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Using relation $v = u + gt \Rightarrow 31.321 = 0 + 9.81 t$

$\therefore t = 3.193 \text{ sec.}$

Ex:2 A stone is dropped into a well and the splash of sound is heard after 90 seconds. Determine the height of drop from the water surface. Assume velocity of sound to be 330 m/sec.

Solution: Let t_1 : time taken by stone to hit the water surface.

t_2 : time taken by sound to travel from water surface till the position of drop. $t_2 = (90 - t_1)$

Let d : depth of water surface.

Sound moves with uniform velocity

consider motion of stone: $u = 0$, $a = 9.81 \text{ m/sec}^2$

Using relation $s = ut + \frac{1}{2}gt^2 = 0 \times t + \frac{1}{2} \times 9.81 t^2$

$\therefore d = 4.905 t_1^2 \dots \dots \textcircled{1}$

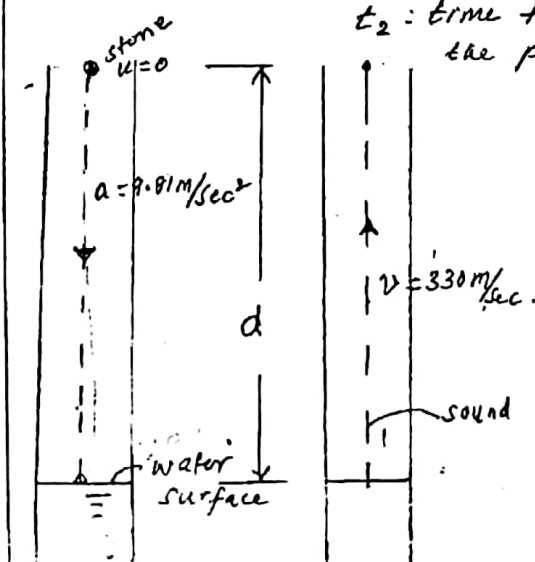
consider motion of sound (uniform velocity)

$\therefore s = vt$ i.e. $d = 330 (90 - t_1) \dots \dots \textcircled{2}$

Equating $\textcircled{1}$ & $\textcircled{2}$ $4.905 t_1^2 = 330 (90 - t_1)$

$\therefore t_1 = 51.13 \text{ sec.}$

depth $= 330 (90 - 51.13) = 12825.78 \text{ m}$



$$v_s^2 - 0 = 2 \times 9.81 \times 37.835$$

$$v_s = 27.25 \text{ m} \quad \text{velocity of strike.}$$

Using relation $v = u + at$

$$\text{Time elapsed from B to C} \quad t_2 = \frac{v - u}{a} = \frac{27.25 - 0}{9.81}$$

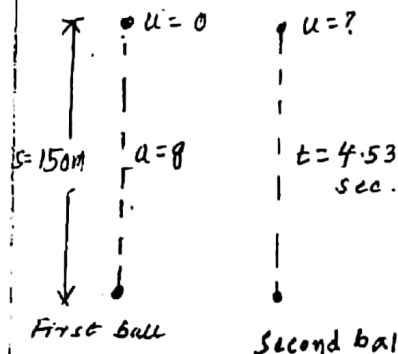
$$t_2 = 2.777 \text{ sec.}$$

Time elapsed from A to B = t_1

$$v = u + at_2 \quad t_2 = \frac{v - u}{a} = \frac{0 - 6}{-9.81} = 0.612 \text{ sec}$$

$$\text{Total time elapsed} = t_1 + t_2 = 3.389 \text{ sec.}$$

Ex: 7 A ball is dropped from the top of a skyscraper building which is 150 m high. After 1 second, another stone is thrown down and strikes the first ball when it has just reached the foot of the building. Find the velocity with which the second ball was thrown.



Consider motion of 1st ball

$$u=0, \quad a=+g=9.81 \text{ m/sec}^2, \quad s=150 \text{ m}, \quad \text{time} = t$$

$$\text{Using } s = ut + \frac{1}{2}at^2$$

$$150 = 0 + \frac{1}{2} \times 9.81 \times t^2$$

$$t = 5.53 \text{ sec.}$$

Consider motion of 2nd ball

$$u=? \quad a=9.81 \text{ m/sec}^2 \quad t = 5.53 - 1 = 4.53 \text{ sec.}$$

$$s = 150 \text{ m}$$

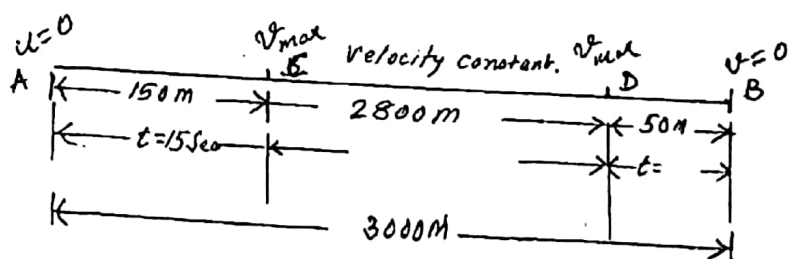
$$\text{Using } s = ut + \frac{1}{2}at^2$$

$$150 = u \times 4.53 + \frac{1}{2} \times 9.81 \times 4.53^2$$

$$\therefore u = \frac{150 - 100.655}{4.53} = 10.893 \text{ m/sec.}$$

Ex: 8 An electric train, travelling between two stations A and B which 3.0 km apart is uniformly accelerated for the first 15 seconds during which period it covers 150 m. It then runs with a constant speed, until it is finally retarded uniformly in the last 50 m. Find the maximum speed of the train and the time taken to complete the journey between the two stations.

Solution:



Consider motion of partical from A to C

$$u=0 \quad v=v_{\text{max}} \quad s=150 \text{ m}$$

$$t=15 \text{ sec.}$$

$$\text{Using } s = ut + \frac{1}{2}at^2$$

$$150 = 0 + \frac{1}{2} \times a \times 15^2$$

$$a = 1.333 \text{ m/sec}^2$$

$$\text{Using relation } v^2 - u^2 = 2as$$

∴ Total time of travel of second ball = $12.232 - 3 = 9.232 \text{ Sec}$.

Time of travel for upward motion = $\frac{9.232}{2} = 4.616 \text{ Sec}$.

Let u_i : Initial velocity of second ball & v : Final velocity

Using relation $v = u + at$

$a = -9.81 \text{ m/s}^2$

$v = 0$

$t = 4.616 \text{ Sec}$

$a = -g$

$$0 = u_i - 9.81 \times 4.616$$

$$\therefore u_i = 45.285 \text{ m/sec}$$

Ex: 5 / A ball was thrown vertically down from a building and it travels a distance of 50m during its 5th second of fall. Find the initial velocity of the body.

A u
 $a = g$
 $t = 4 \text{ Sec}$

B 4th Sec. end

50m $t = 1 \text{ Sec}$

$a = g$
C 5th second end

consider motion of particle in 5th second.

distance travelled = 50m time = 1 Sec $a = 9.81 \text{ m/s}^2$

Let v : final velocity at the end of 5th second

u_f : Initial velocity at the beginning of 5th second

Using relation $s = ut + \frac{1}{2}at^2$

$$50 = u_f \times 1 + \frac{1}{2} \times 9.81 \times 1^2$$

$$\therefore u_f = 50 - \frac{9.81}{2} = 45.095 \text{ m/sec}$$

consider motion of particle from A to B

Initial velocity = u , final velocity = $v_f = 45.095 \text{ m/s}$

time = $t = 4 \text{ Sec}$ $a = +g$

Using relation $v = u + at$

$$45.095 = u + 9.81 \times 4$$

$$u = 5.855 \text{ m/sec}$$

Ex: 6 / A lift is going up with a velocity of 6m/sec. when the lift was 36m above the bottom of the shaft, a bolt gets detached from the bottom of the lift floor. Find the velocity with which the bolt strikes the bottom of the shaft and the time elapsed.

Consider motion of bolt: When the bolt gets detached it moves

in upward direction with initial velocity $u = 6 \text{ m/s}$ at A till B where its velocity becomes zero.

Consider motion of bolt from A to B.

$u = 6 \text{ m/sec}$ $v = 0$ $a = -9.81 \text{ m/sec}^2$

Let s_1 : distance from A to B.

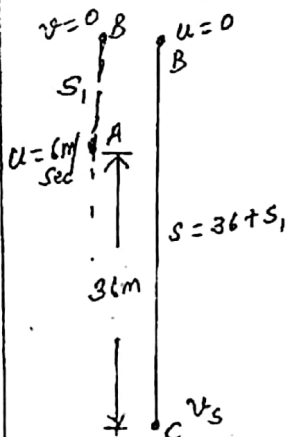
Using $v^2 - u^2 = 2as \Rightarrow 0 - 6^2 = 2 \times (-9.81) \times s_1$

$$\therefore s_1 = 1.835 \text{ m}$$

Consider motion of bolt from B to C

$u = 0$ $v = v_s$ $a = +g = 9.81 \text{ m/sec}^2$ $s = 36 + 1.835 = 37.835$

Using relation $v^2 - u^2 = 2as$



Consider motion of train from C to D. : Train moves with uniform velocity

$$S = 2800 \text{ m} \quad v = v_{\text{max}} = 19.99 \text{ m/sec} \quad t = t_2$$

using $S = vt$

$$t_2 = \frac{S}{v} = \frac{2800}{19.99} = 140.07 \text{ Sec.}$$

Consider motion of train from D to C : Train moves with uniform retardation

$$u = v_{\text{max}} = 19.99 \text{ m/sec} \quad v = 0 \quad S = 50 \text{ m} \quad \text{time of travel} = t_3$$

using $v^2 - u^2 = 2as$

$$0 = 19.99^2 = 2a_2 \times 50$$

$$a = a_2$$

$$\therefore a_2 = -7.99 \text{ m/sec}^2 \quad (-\text{ve sign indicates retardation})$$

using $v = u + at$

$$0 = 19.99 + (-7.99) t_3$$

$$\therefore t_3 = \frac{19.99}{7.99} = 2.50 \text{ Sec.}$$

$$\text{Total time} = t_1 + t_2 + t_3 = 15 + 140.07 + 2.50 = 157.57 \text{ Sec.}$$

$$\text{Max Velocity} = 19.99 \text{ m/sec} = \frac{19.99 \times 3600}{1000} = 71.964 \text{ km/hour}$$

Non uniform motion of a body:-

A body is said to be in non uniform motion, when the velocity or acceleration are not same at regular intervals of time.

Mathematically, the velocity

$$v = \frac{ds}{dt}$$

and acceleration,

$$a = v \times \frac{dv}{ds}$$

Methods of analysing non-uniform motion:-

- i) By differentiation.
- ii) By Integration.

i) Determination of velocity and acceleration by differentiation

The equation of motion is some times expressed as,

$$s = 4t^3 + 8t^2 + 7t + 6 \quad \text{--- (1) and}$$

asked to determine velocity and

acceleration at some particular interval of time 't'.

To get velocity:

Differentiate the above equation and substitute the value of 't'.

$$\frac{ds}{dt} = 12t^2 + 16t + 7 = v \quad \dots (ii)$$

To get acceleration:-

Differentiate the equation (ii) and substitute the value of 't'.

$$\frac{dv}{dt} = 24t + 16 \quad \dots (iii)$$

⇒ A particle starting from rest, moves in a straight line, whose equation of motion is given by $s = 5t^3 - 3t^2 + 6$. Find the displacement, velocity, and acceleration of the particle after 5 seconds.

Solution:-

$$S = 5t^3 - 3t^2 + 6$$

Displacement $S = 5 \times 5^3 + 3 \times 5^2 + 6 = \underline{556\text{m}}$

Velocity:-

$$\frac{dS}{dt} = v = 15t^2 - 6t$$

Now substitute $t = 5\text{ seconds}$

$$v = 15 \times 5^2 - 6 \times 5 = 345\text{m/sec.}$$

Acceleration:-

$$a = \frac{dv}{dt} = 30t - 6$$

put $t = 5\text{ sec.}$

$$a = 30 \times 5 - 6 \\ = \underline{144\text{m/sec}^2}$$

- 2) A train moves along a straight line whose equation of motion is given by $S = 15t + 5t^2 - 3t^3$ where S in m and t is in seconds. Determine i) Velocity and acceleration at start ($t=0$)
ii) Acceleration when velocity is zero.

Solution:-

Equation for velocity.

$$\frac{ds}{dt} = 15 + 10t - 9t^2 = v \dots (1)$$

Equation for acceleration

$$a = \frac{dv}{dt} = 10 - 18t \dots (2)$$

i) Velocity and acceleration when
 $t = 0$

$$v = 15 + 10 \times 0 - 9(0)^2 \\ = 15 \text{ m/sec}$$

$$a = 10 - 18 \times 0 = 10 \text{ m/sec}^2$$

ii) Acceleration when velocity is
zero, i.e. $v = 0$

$$0 = 15 + 10t - 9t^2 \\ 0 = 15 + 10t - 9t^2 \\ t = 1.96 \text{ Sec.}$$

$$\therefore a = 10 - 18 \times 1.96 = -25.28 \text{ m/sec}^2 \\ (\text{retardation})$$

3) A vehicle is moving with variable acceleration and its motion is given by the equation, $S = 25t + 4t^2 - 6t^3$, where 'S' is in m and t is in seconds. Determine.

- i) Velocity and acceleration at start.
- ii) The time when the vehicle reaches its maximum velocity.
- iii) The maximum velocity of the vehicle.

Solution:-

Equation of distance

$$S = 25t + 4t^2 - 6t^3 \quad \text{--- (1)}$$

$$\text{Velocity } v = \frac{ds}{dt} = 25 + 8t - 18t^2 \quad \text{--- (2)}$$

$$\text{Acceleration } a = \frac{dv}{dt} = 8 - 36t \quad \text{--- (3)}$$

i) Velocity and acceleration at start
ie $t = 0$

$$v = 25 \text{ m/s}$$

$$a = 8 \text{ m/s}^2$$

ii) The time t when the vehicle reaches its maximum velocity.

$$\text{For max. velocity } \frac{dv}{dt} = 0$$

$$8 - 36t = 0$$

$$t = 0.222 \text{ s}$$

iii) Maximum velocity

$$\begin{aligned} v_{\max} &= 25 + 8(0.222) - 18(0.222)^2 \\ &= 26.776 - 0.887 \\ &= \underline{\underline{25.88 \text{ m/s}}} \end{aligned}$$

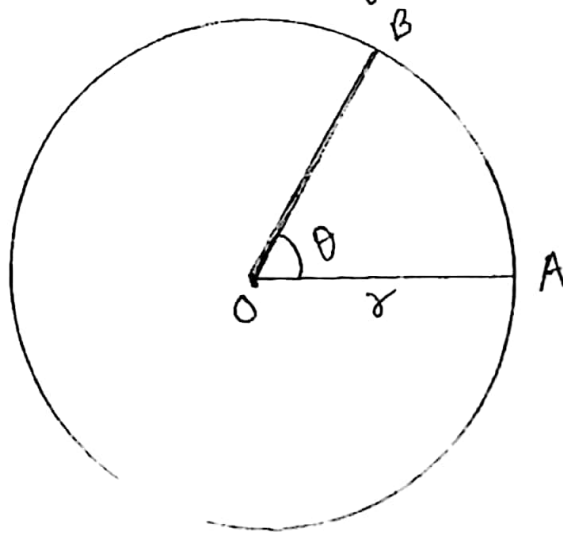
Curvilinear motion of Particles

A particle is said to possess a curvilinear motion when it traces a curved path. This curved path may be a simple circular path or any other curve like open curve or closed curve.

Definitions:-

i) Angular Displacement θ :- The total angle traced by a particle with respect to its original position along circular path is called angular displacement (θ), measured in radians.

ii) Angular velocity:-



It is defined as the rate of change of angular displacement (θ) with respect to time t . It is expressed by ' ω ' as radians/sec.

From fig,

$$\text{Angular velocity} = \frac{\text{Angular displacement}}{\text{time}}$$

$$\omega = \frac{\theta}{t}$$

$$\text{or } \omega = \frac{d\theta}{dt} \text{ mathematically.}$$

The relation between angular velocity ' ω ' and linear velocity ' v ' is given by,

$$v = r \omega$$

$$= \text{Radius} \times \text{Angular velocity}$$

Let ' T ' be the time required for one revolution then angle traversed

$$\theta = 2\pi$$

$$\omega = \frac{2\pi}{T}$$

$$v = \frac{2\pi r}{T}$$

Angular Acceleration:-

It is defined as the rate of change of angular velocity. It is measured in radians/sec². It is expressed as α .

$$\begin{aligned}\therefore \alpha &= \text{Rate of change of angular velocity} \\ &= \frac{d\omega}{dt} = \frac{d}{dt} \left(\frac{d\theta}{dt} \right) \quad (\because \omega = \frac{d\theta}{dt}) \\ &= \frac{d^2\theta}{dt^2}\end{aligned}$$

$$\text{Also } \frac{d\omega}{dt} = \frac{d\omega}{d\theta} \times \frac{d\theta}{dt} = \frac{d\omega}{d\theta} \times \omega \quad (\because \omega = \frac{d\theta}{dt})$$

But $\frac{d\omega}{dt}$ is angular acceleration ' α '.

$$\therefore \alpha = \frac{d\omega}{dt} = \omega \times \frac{d\omega}{d\theta}$$

$$\boxed{\therefore \alpha = \omega \frac{d\omega}{d\theta}}$$

Relation between linear Acceleration and angular acceleration:-

Linear acceleration = Radius $\times$ Angular acceleration.

$$a = r \times \alpha$$

Equations of motion along circular path:-

$$i) \omega = \omega_0 + \alpha t$$

Where ω = Final angular velocity
 ω_0 = Initial angular velocity
 α = Angular acceleration.

$$ii) \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

θ = Angular displacement.

$$iii) \omega^2 - \omega_0^2 = 2\alpha\theta.$$

Relation between r.p.m (N) and angular velocity is:-

Let N be the number of revolutions per minute.

Angle covered by a body in one second = Angle covered in one revolution $\times$ No. of revolutions per second.

$$= 2\pi \times \frac{N}{60}$$

But angle covered by the body per second is angular velocity is

$$\therefore \omega = \frac{2\pi N}{60}$$

$$\text{Linear speed} = v = \omega r = \frac{2\pi N}{60} \times r$$

$$v = \frac{\pi d N}{60}$$

Where d = Dia of the path.

Problem:-

→ A body is rotating with an angular velocity of 6 rad/sec . After 5 seconds the angular velocity becomes 15 rad/sec . Determine angular acceleration of the body.

Solution:-

Initial angular velocity $\omega_0 = 6 \text{ rad/sec}$

Final angular velocity $\omega = 15 \text{ rad/sec}$

Time taken $t = 5 \text{ sec}$

We have $\omega = \omega_0 + \alpha t$

$$15 = 6 + \alpha \times 5$$

$$\alpha = \underline{1.8 \text{ rad/sec}^2} \text{ (Angular acceleration)}$$

ation)

2) A wheel is rotating about a fixed axis at 20 rpm. It is uniformly accelerated for 70 seconds, during this time it makes 50 revolutions. Determine
 i) Angular velocity at the end of this interval. ii) Time required for the speed to reach 110 rpm.

Solution

We have ^{initial} angular velocity,

$$\omega_0 = \frac{2\pi N}{60}$$

$$= \frac{2\pi \times 20}{60} = \underline{2.0948 \text{ rad/sec.}}$$

Since the wheel rotates for 50 revolutions the angular displacement,

$$= 2\pi \times \text{No. of rotations}$$

$$\theta = 2\pi \times 50 = \underline{100\pi}$$

Also we have

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$100\pi = 2.094 + \frac{1}{2} \alpha \times 70^2$$

$$\alpha = \underline{0.0688 \text{ rad/sec}^2}$$

3) A flywheel starts rotating from rest and is given an angular acceleration of 1 rad/sec^2 . Determine the angular velocity and speed in rpm after 90 seconds. If the flywheel is brought to rest with an uniform angular retardation of 0.5 rad/sec^2 , find the time required by the flywheel to come to rest.

Solution:-

Initial angular velocity $\omega_0 = 0$

Angular acceleration $= 1 \text{ rad/sec}^2$.

ω be the final angular velocity after 90 sec.

$$\begin{aligned} \text{Now } \omega &= \omega_0 + \alpha t \\ &= 0 + 1 \times 90 = \underline{90 \text{ rad/sec}} \end{aligned}$$

$$\text{Also } \omega = \frac{2\pi N}{60}$$

$$90 = \frac{2 \times \pi \times N}{60}$$

$$N = \underline{\underline{859.44 \text{ rpm}}}$$

't' is the time taken to bring the flywheel

$$\text{Let } \omega_0 = 90 \text{ rad/sec.}$$

$$\alpha = -0.5 \text{ rad/sec}$$

$$\omega = 0$$

$$\omega = \omega_0 + \alpha t$$

$$= 90 - 0.5 \times t$$

$$t = \underline{\underline{180 \text{ seconds.}}}$$

4) A flywheel rotates at 200 rpm. A uniform retardation is given for 10 seconds and rotation reduces to 160 rpm.

i) Determine the number of revolutions made by the flywheel before it comes to rest.

ii) Time taken by flywheel before it comes to rest from 200 rpm.

Solution:- case (i)

Initial rotation $N_0 = 200 \text{ rpm}$

∴ Initial angular velocity

$$\begin{aligned}\omega_0 &= \frac{2\pi N}{60} \\ &= \frac{2 \times \pi \times 200}{60} = 20.948 \text{ rad/sec}\end{aligned}$$

Final velocity when rotation reduces to

1608 rpm

$$\omega = \frac{2\pi N}{60} = \frac{2 \times \pi \times 160}{60} = 16.768 \text{ rad/sec}$$

Time of retardation $t = 10 \text{ sec}$.

From relation

$$\omega = \omega_0 + \alpha t$$

$$16.76 = 20.94 + \alpha \times 10$$

$$\alpha = -0.418 \text{ rad/sec (Retardation)}$$

Let θ be the displacement before it comes to rest from 2008 rpm.

Initial velocity $\omega_0 = 20.948 \text{ rad/sec}$.

Final velocity $\omega = 0$

$$\omega^2 - \omega_0^2 = 2 \times \alpha \times \theta$$

$$0^2 - 20.94^2 = 2 \times (-0.418) \theta$$

$$\theta = \underline{\underline{524.58 \text{ rad}}}$$

∴ We have

$$\theta = 2\pi \times N$$

$$N = \frac{\theta}{2\pi} = \frac{524.5}{2\pi} = \underline{\underline{83.48 \text{ revolutions}}}$$

Case(ii) Time taken before it comes to rest
from 200 rpm

$$\omega = \omega_0 + \alpha t$$

$$0 = 20.94 + (-0.418) t$$

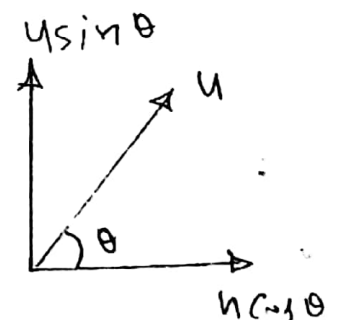
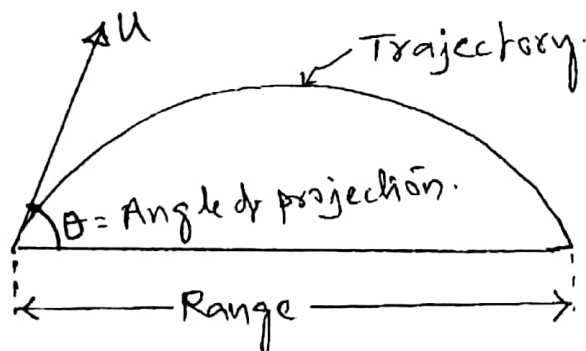
$$t = \underline{\underline{50 \text{ sec}}}$$

Projectiles:-

Introduction:- In case of rectilinear motion, we studied the motion of a particle either in horizontal or in vertical direction only.

But when a particle is projected in air at some particular angle (other than 90° w.r. to horizontal) it is subjected to both horizontal and vertical components of velocity. The horizontal component of velocity remains constant and impart the horizontal motion to the particle. The vertical component of velocity is affected by gravitational force of earth. Hence the path traced by the particle is parabolic.

Terminology:-



Trajectory:- The path traced by the projectile in the space is known as trajectory.

Velocity of projection (u):- The velocity with which the particle is projected in to space is called velocity of projection.

Angle of projection (θ):- The angle made by velocity of projection with respect to horizontal is called angle of projection.

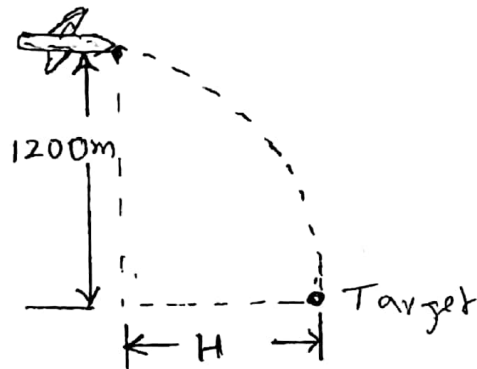
Time of flight (T):- The total time a particle remained in space from the instant of its projection till it reaches the ground is called time of flight.

Range (R):- The horizontal distance from the point of projection of a particle till reaches the ground is called range.

Prob 1 $\rightarrow$ An aircraft moving horizontally at 120 km/hr speed at an elevation of 1200 m targets a point on the ground and releases a bomb which hits it. Determine the horizontal distance of the aircraft (position

when it releases the bomb from the target. Also calculate the velocity and direction with which bomb hits the target.

Solution:-



Solution:-

Consider vertical motion of the bomb

Initial velocity in vertical direction

$$u_v = 0$$

$$\text{Distance } s = 1200\text{m}$$

Time taken by bomb to hit target = 't'

$$\text{Using relation } s = ut + \frac{1}{2}gt^2$$

$$1200 = 0 \times t + \frac{1}{2} \times 9.81 \times t^2$$

$$t = \underline{\underline{15.64 \text{ Sec}}}$$

Horizontal velocity remaining constant

$$v_h = 120 \text{ km/hr} = 120 \times 1000 / 60 \times 60 = \underline{\underline{33.33 \text{ m/sec}}}$$

Horizontal distance of target = $H = v_h \times t$

$$= 33.33 \times 15.64$$

$$= 521.33 \text{ m}$$

From vertical motion,

$u_v = 0$ v_v = Vertical velocity when it hits the target.

$$t = 15.64 \text{ sec} \quad g = 9.81 \text{ m/sec}^2$$

We have

$$v = u + gt$$

$$v_v = 0 + 9.81 \times 15.64$$
$$= \underline{153.43 \text{ m/sec}}$$

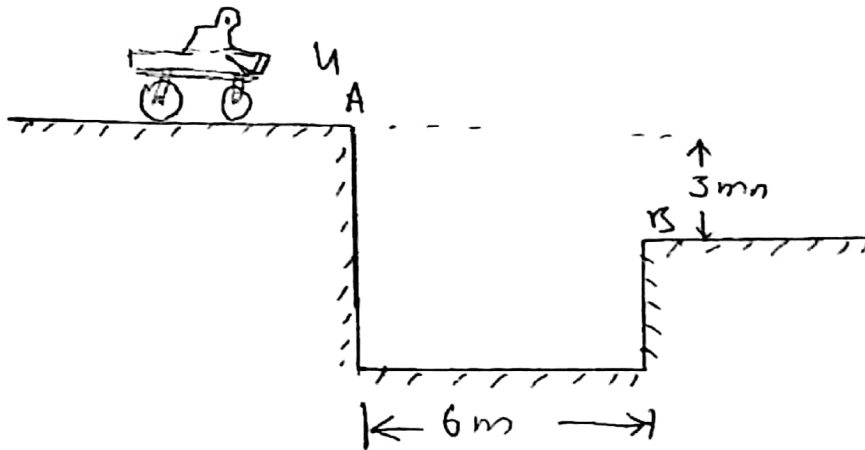
$$\therefore \text{Velocity of hit} = \sqrt{u_h^2 + v_v^2}$$
$$= \sqrt{33.33^2 + 153.43^2}$$
$$= \underline{157.01 \text{ m/sec}}$$

Direction or angle of hit

$$\tan \theta = \frac{v_v}{u_h} = \frac{153.43}{33.33}$$

$$\theta = \underline{77.74^\circ}$$

2) A bullet motor cyclist wants to jump over a ditch as shown in fig. Determine necessary minimum velocity 'u' in kmph. Also determine the inclination and the magnitude of the velocity of the motor cyclist after clearing the ditch.



Solution:-

$$H = 6\text{m}$$

$$\text{Vertical distance} = S_v = 3\text{m}$$

Initial velocity in vertical direction

$$u_v = 0$$

$$a = g = 9.81\text{ m/sec}^2$$

t = time of travel from A to B.

v_v = Final velocity.

$$\text{Using } S_v = u_v t + \frac{1}{2} g t^2$$

$$3 = 0 \times t + \frac{1}{2} \times 9.81 \times t^2$$

$$t = \underline{\underline{0.782\text{ Sec}}}$$

$$\text{Also } v_v = u + g t$$

$$v_v = 0 + 9.81 \times 0.782$$

$$= \underline{\underline{7.672\text{ m/sec}}}$$

Considering horizontal motion:-

Horizontal velocity is constant = v_h

Using $s = v_h \times t$

$$6 = v_h \times 0.782$$

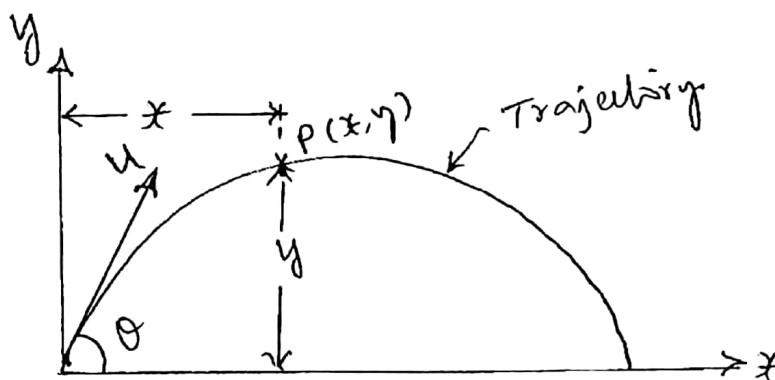
$$v_h = \underline{7.673 \text{ m/s}}$$

$$\therefore \text{Resultant velocity at B} = \sqrt{7.672^2 + 6.672^2} \\ = \underline{10.857 \text{ m/s}}$$

$$\text{ inclination } \theta = \tan^{-1} \frac{v_v}{v_h} = \underline{45^\circ}$$

= =

Equation for the path of a projectile:-



$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$$

Time of flight:

$$t = \frac{2u \sin \theta}{g}$$

Horizontal Range:-

$$R = \frac{u^2 \times \sin 2\theta}{g}$$

Maximum Horizontal Range:-

$$R_{\max} = \frac{u^2}{g}$$

Maximum height of projectile

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

Prob 1 A particle is projected at an angle of 50° with the horizontal. The horizontal range of the particle is 6 km. Find: i) The velocity of projection. ii) The maximum height reached by the projectile.

Solution:-

Given $\theta = 50^\circ$

$R = 6 \text{ km} = 6000 \text{ m.}$

i) Using $R = \frac{u^2 \sin 2\theta}{g}$

$$6000 = \frac{u^2 \sin 2 \times 50}{9.81}$$

$$u = \underline{\underline{244.48 \text{ m/sec}}}$$

ii) Maximum height attained by the particle.

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{244.48^2 \sin^2 50}{2 \times 9.81}$$

$$= \underline{\underline{1787.7 \text{ m}}}$$

Prob 2: A particle is projected in air with a velocity of 120 m/sec at an angle of 30° with horizontal. Determine i) the horizontal range ii) The maximum height by the particle iii) The time of flight.

Solution:-

Velocity of projection $u = 120 \text{ m/sec}$

Angle of projection $\theta = 30^\circ$

i) Horizontal range

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{120^2 \times \sin 2 \times 30}{9.81} \\ = \underline{\underline{1271.23 \text{ m}}}$$

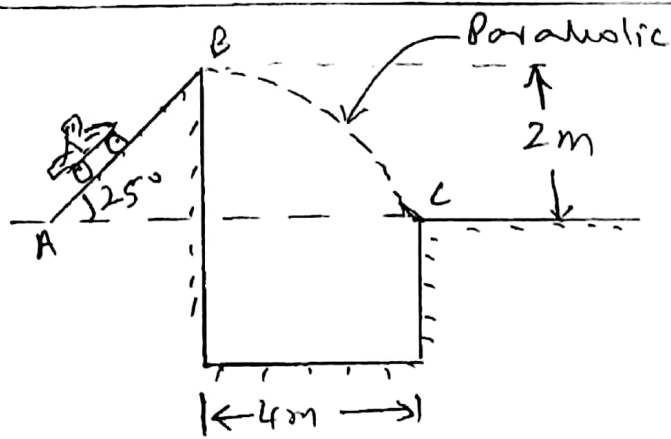
ii) Maximum height

$$H_{\text{max}} = \frac{u^2 \sin^2 \theta}{2g} \\ = \frac{120^2 \times \sin^2 30}{2 \times 9.81} = \underline{\underline{183.49 \text{ m}}}$$

iii) Time of flight

$$T = \frac{2u \sin \theta}{g} = \frac{2 \times 120 \times \sin 30}{9.81} \\ = \underline{\underline{12.23 \text{ sec}}}$$

- ③ A motor cyclist wants to clear a ditch shown in fig. If the ramp at B has a slope of 25° , determine the minimum speed of the motor cycle to touch the point C.



Solution:- Angle of ramp = $\theta = 25^\circ$.

ve distance = -2m (Down to 'B')

H^e Range $R = 4\text{m}$

The path traced by the motor cycle from B to C is parabolic.

We have

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

Here, $y = -2\text{m}$

$x = 4\text{m}$

$$-2.0 = 4 \tan 25 - \frac{9.81(4)^2}{2u^2 \cos^2 25}$$

$$u = \underline{\underline{4.97\text{m/s}}}$$

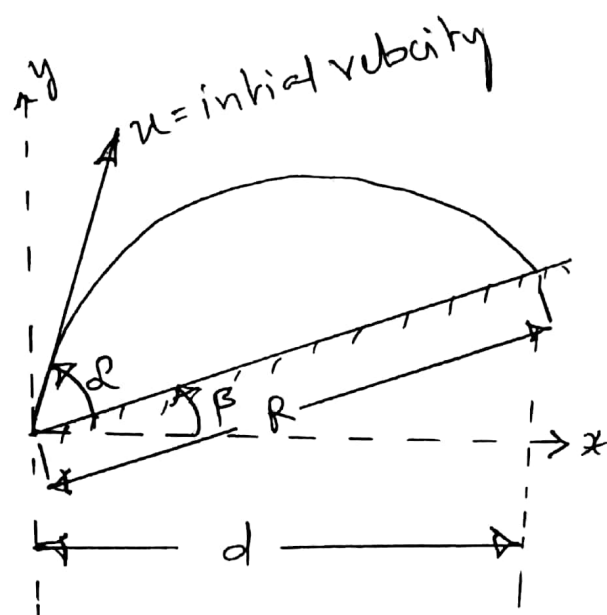
Super elevation:-

To avoid skidding and overturning the outer edge of the pavement on highway curves and outer rail on railway curves are elevated above the inner edge or inner rail. This raising of outer edge on outer rail above the inner edge or inner rail by certain amount is called as super elevation.

Following are the advantages of providing super elevation:-

- i) Sliding, skidding and overturning is avoided.
- ii) Higher speed can be permitted on curved path.
- iii) Lateral pressure may be eliminated giving more comfort to passengers.
- iv) Wear and tear of wheels is reduced.

Projection on inclined plane:-



i) Range of inclined plane (R)

$$R = \frac{2u^2 \cos \alpha \sin(\alpha - \beta)}{g \cos^2 \beta} \quad \text{--- (1)}$$

Maximum range

$$R_{\max} = \frac{u^2}{g(1 + \sin \beta)}$$

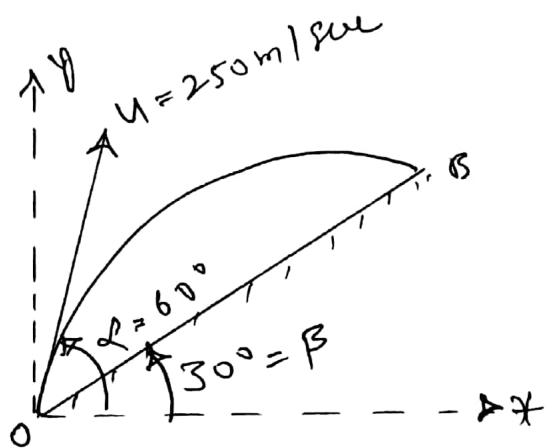
ii) Time of flight (t)

$$t = \frac{2u \sin(\alpha - \beta)}{g \cos \beta} \quad \text{--- (2)}$$

[Note: If the plane is inclined downward then the range and time of flight could be obtained by substituting $\beta = -\beta$ in eqn (1) & (2)]

Prob 1 → A projectile is projected with a velocity of 250 m/sec at an angle of 60° with horizontal from the foot of a plane having inclination 30° . Determine the distance where the projectile will strike the plane (range) and the time of flight taken. Take $g = 9.81 \text{ m/sec}^2$.

Solution:-



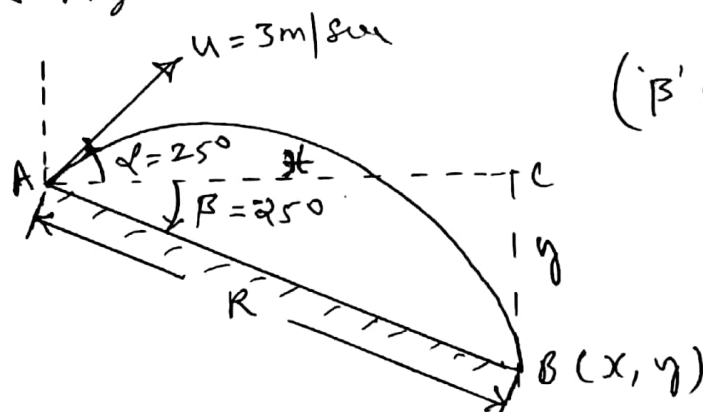
Given, $\alpha = 60^\circ$, $\beta = 30^\circ$, $u = 250 \text{ m/sec}$

Distance where the projectile strikes the plane, Range

$$\begin{aligned}
 R &= \frac{2u^2 \cos \alpha [\sin(\alpha - \beta)]}{g \cos^2 \beta} \\
 &= \frac{2 \times 250^2 \times \cos 60 [\sin(60 - 30)]}{9.81 \cos^2 30} \\
 &= \underline{\underline{4247.37 \text{ m}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Time of flight } t &= \frac{2u}{g} \frac{\sin(\alpha - \beta)}{\cos \beta} \\
 &= \frac{2 \times 250}{9.81} \frac{\sin(60 - 30)}{\cos 30} \\
 &= \underline{\underline{29.427 \text{ Sec}}}
 \end{aligned}$$

2) A stone is projected from A located on a slope with a velocity 3m/sec. at an angle 25° with the horizontal as shown. Determine Range, time of flight and co-ordinates of the point B at which it will hit the slope whose inclination is 25° to the horizontal.



Solution:- $u = 3 \text{ m/sec}$, $\alpha = 25^\circ$, $\beta = -25^\circ$
(Inclined downwards)

$$\begin{aligned}
 \text{Range } R &= \frac{2u^2 \cos \alpha \sin(\alpha - \beta)}{g \cos^2 \beta} \\
 &= \frac{2(3)^2 \cos 25 \sin(25 - (-25))}{9.81 \cos^2(25)} = \underline{\underline{1.55 \text{ m}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Time of flight } t &= \frac{2u}{g} \frac{\sin(\alpha - \beta)}{\cos \beta} \\
 &= \frac{2 \times 3}{9.81} \frac{\sin(25 - (-25))}{\cos(-25)} \\
 &= \underline{\underline{\text{Seconds}}}
 \end{aligned}$$

Let (x, y) be the co-ordinates of B.

From $\triangle ABC$

$$\tan 25 = \frac{y}{x}$$

$$y = 0.466x$$

$$\text{Also } R^2 = x^2 + y^2$$

$$1.55^2 = x^2 + (0.466x)^2$$

$$x = 1.274$$

$$\begin{aligned}
 \therefore y &= 0.466 \times x = 0.466 \times 1.274 \\
 &= \underline{\underline{0.594\text{m}}}
 \end{aligned}$$

$\therefore$ co-ordinates of B $(1.274, 0.594)$

D'Alembert's principle:-

The principle states that the resultant of the external forces and the kinetic reaction acting on a body equals to zero. The kinetic reaction is defined as the negative of the product of the mass m and acceleration a . The principle therefore states as

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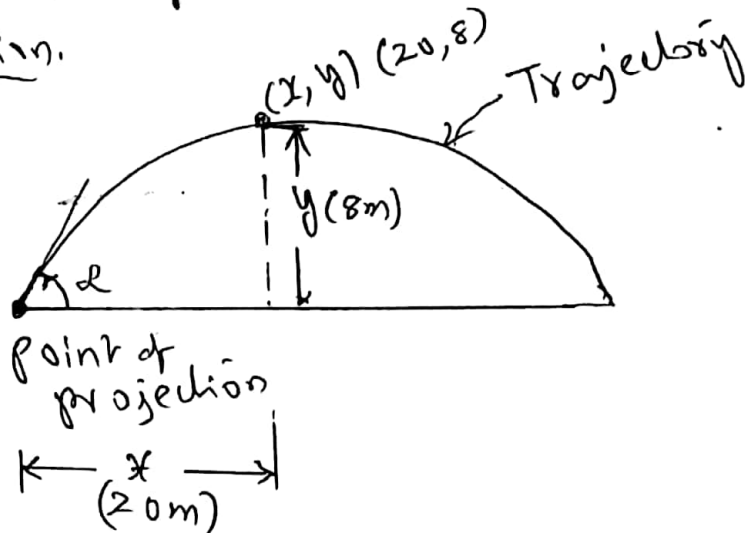
Applications: i) The technique used in solving static problems may then provide relatively simple solutions to some problems in dynamics.

ii) D'Alembert's principle is especially useful in problems involving constraints.

Model Q.8 problem:-

→ A particle is projected with a velocity of 20 m/sec . in air at angle α with the horizontal. The 'x' and y co-ordinates of a point lying on the trajectory of the particle with respect to point of projection are 20 m and 8 m respectively. Find the angle of projection of the particle.

Solution.



Velocity of projection $u = 20\text{ m/sec}$

Angle of projection $= \alpha$

$$x = 20\text{ m}$$

$$y = 8\text{ m}$$

We have equation for path traced by a projectile

$$y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha}$$

$$8 = 20 \tan \alpha - \frac{9.81 \times 20^2}{2 \times 20^2 \cos^2 \alpha}$$

$$= 20 \tan \alpha - \frac{4.905}{\cos^2 \alpha} \left[\frac{1}{\cos^2 \alpha} = \sec^2 \alpha \right]$$

$$= 20 \tan \alpha - 4.905 \sec^2 \alpha$$

$$\sec^2 \alpha = [1 + \tan^2 \alpha]$$

$$= 20 \tan \alpha - 4.905 (1 + \tan^2 \alpha)$$

$$4.905 \tan^2 \alpha - 20 \tan \alpha + 4.905 + 8 = 0$$

$$4.905 \tan^2 \alpha - 20 \tan \alpha + 12.905 = 0$$

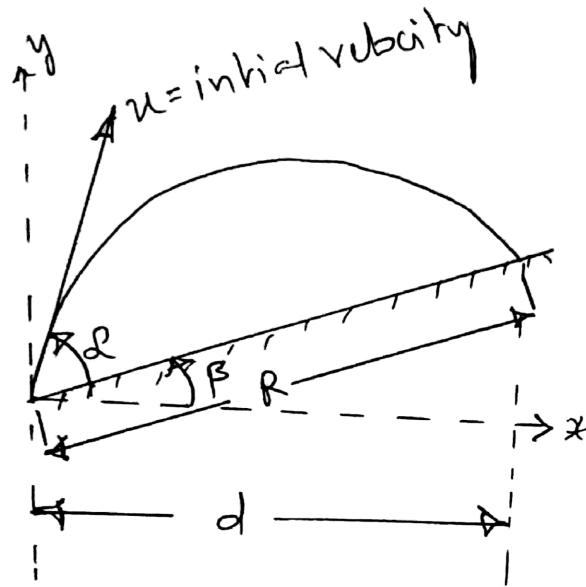
$$\tan \alpha = \frac{-20 \pm \sqrt{(20)^2 - 4 \times 4.905 \times 12.905}}{2 \times 4.905}$$

$$\tan \alpha = 3.273 \text{ and } \alpha = \underline{73^\circ 0.8'}$$

$$\tan \alpha = 0.8036 \text{ and } \alpha = \underline{38^\circ 47'}$$

Hence for the given problem there are two angles of projection.

Projection on inclined plane:-



i) Range of inclined plane (R)

$$R = \frac{2u^2 \sin \alpha \cos(\alpha - \beta)}{g \cos^2 \beta} \quad \text{--- (1)}$$

Maximum range

$$R_{\max} = \frac{u^2}{g(1 + \sin \beta)}$$

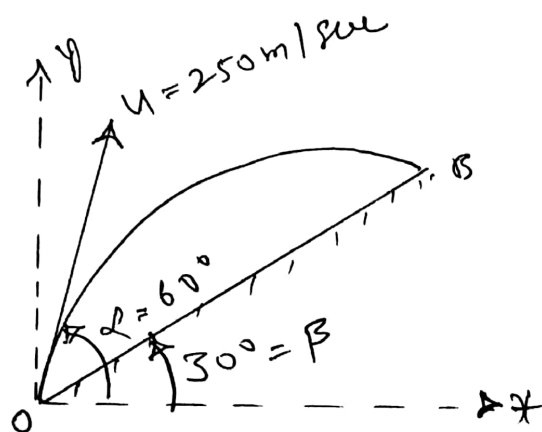
ii) Time of flight (t)

$$t = \frac{2u \sin(\alpha - \beta)}{g \cos \beta} \quad \text{--- (2)}$$

[Note: If the plane is inclined downward then the range and time of flight could be obtained by substituting $\beta = -\beta$ in eqn (1) & (2)]

Prob 1 → A projectile is projected with a velocity of 250 m/sec at an angle of 60° with horizontal from the foot of a plane having inclination 30° . Determine the distance where the projectile will strike the plane (range) and the time of flight taken. Take $g = 9.81 \text{ m/sec}^2$.

Solution:-



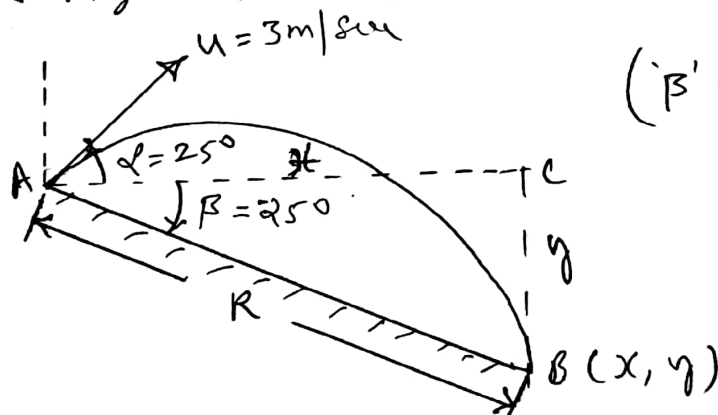
Given, $\alpha = 60^\circ$, $\beta = 30^\circ$, $u = 250 \text{ m/sec}$

Distance where the projectile strikes the plane, Range

$$\begin{aligned}
 R &= \frac{2u^2 \cos \alpha [\sin(\alpha - \beta)]}{g \cos^2 \beta} \\
 &= \frac{2 \times 250^2 \times \cos 60 [\sin(60 - 30)]}{9.81 \cos^2 30} \\
 &= \underline{\underline{4247.37 \text{ m}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Time of flight } t &= \frac{2u}{g} \frac{\sin(\alpha - \beta)}{\cos \beta} \\
 &= \frac{2 \times 250}{9.81} \frac{\sin(60 - 30)}{\cos 30} \\
 &= \underline{\underline{29.427 \text{ Sec}}}
 \end{aligned}$$

2) A stone is projected from A located on a slope with a velocity 3m/sec. at an angle 25° with the horizontal as shown. Determine Range, time of flight and co-ordinates of the point B at which it will hit the slope whose inclination is 25° to the horizontal.



('β' should be '-')

Solution:- $u = 3 \text{ m/sec}$, $\alpha = 25^\circ$, $\beta = -25^\circ$
(Inclined downwards)

$$\begin{aligned}
 \text{Range } R &= \frac{2u^2 \cos \alpha \sin(\alpha - \beta)}{g \cos^2 \beta} \\
 &= \frac{2(3)^2 \cos 25^\circ \sin(25^\circ - (-25^\circ))}{9 \cos^2(25^\circ)} = \underline{\underline{1.55 \text{ m}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Time of flight } t &= \frac{2u}{g} \frac{\sin(\alpha - \beta)}{\cos \beta} \\
 &= \frac{2 \times 3}{9.81} \frac{\sin(25 - (-25))}{\cos(-25)} \\
 &= \underline{\underline{\hspace{2cm}}} \text{ Seconds}
 \end{aligned}$$

Let (x, y) be the co-ordinates of B.

From $\triangle ABC$

$$\tan 25 = \frac{y}{x}$$

$$y = 0.466x$$

$$\text{Also } R^2 = x^2 + y^2$$

$$1.55^2 = x^2 + (0.466x)^2$$

$$x = 1.274$$

$$\begin{aligned}
 \therefore y &= 0.466 \times x = 0.466 \times 1.274 \\
 &= \underline{\underline{0.594\text{m}}}
 \end{aligned}$$

$\therefore$ Co-ordinates of B $(1.274, 0.594)$

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The principle states that the resultant of the external forces and the kinetic reaction acting on a body equals to zero. The kinetic reaction is defined as the negative of the product of the mass m and acceleration a . The principle therefore states as

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